

Lab Report

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Time Spent: 00:11

Score: 0/2 (0%)

Not Passed

Passing Score: 2/2 (100%)

TASK SUMMARY

Lab Questions

Filter for SYN and ACK packets

Q1 Which sign indicates that this is a DISTRIBUTED denial-of-service (DDoS) attack?

Your answer:

Correct answer: There are multiple source addresses for the SYN packets with the destination address 198.28.1.1.

EXPLANATION

Complete this lab as follows:

1. Use Wireshark to capture packets and filter for packets with the SYN flag set.
 - a. From the Favorites bar, select **Wireshark**.
 - b. Under Capture, select **enp2s0**.
 - c. From the menu, select the **blue fin** to start the capture.
 - d. In the *Apply a display filter* field, type **tcp.flags.syn==1 and tcp.flags.ack==0** and press **Enter** to filter the Wireshark display to show packets with only the SYN flag.

Notice that there is a flood of SYN packets being sent to 198.28.1.1 (www.corpnet.xyz).

2. Use a filter to display only packets that have the SYN flag and the ACK flag set.
 - a. In the *Apply a display filter* field, change the ending of the **tcp.flags.ack** portion from a **0** to a **1** and press **Enter** to filter the Wireshark display to only those packets with both the SYN flag and ACK flag.

You should notice that there are far fewer SYN-ACK packets than SYN packets. The server is so busy that it can't respond to all of the packets.

- b. Select the **red square** to stop the capture.
3. Use a filter to only display packets that contain the ACK flag and answer the question.
 - a. In the *Apply a display filter* field, change the ending of the **tcp.flags.syn** portion from a **1** to a **0** and press **Enter** to filter the Wireshark display to packets with only the ACK flag.

You should see ACK packets, but none of them are being sent to 198.28.1.1 (www.corpnet.xyz). In a SYN attack, the ACK packets are never sent. This ties up the half-open connections on the server.

- b. In the top right, select **Answer Questions**.
 - c. Answer the question.
 - d. Select **Score Lab**.